

Day 18 Problem Set

Geometry Summer Credit | Circle Segment Relationships

Name:	Date:	Period:
-------	-------	---------

Directions

Name the configuration and write its product rule before solving. For secants, label external, internal, and whole lengths.

Part A: Tangents and Chords

1.	A radius meets a tangent at the point of tangency. Find the angle formed.
2.	Two tangent segments from one external point are $2x+7$ and $4x-9$. Find x and each length.
3.	Two tangent segments from one external point are $5x-2$ and $3x+14$. Find x and each length.
4.	Intersecting chord parts are 4 and 15 on one chord and 6 and x on the other. Find x .
5.	Intersecting chord parts are 9 and 8 on one chord and 12 and x on the other. Find x .
6.	Intersecting chord parts are $x+2$ and 6 on one chord and 4 and 12 on the other. Find x .

Part B: Secants and Tangent-Secants

7. Two secants share an external point. First: external 4, whole 18. Second: external 6, whole x . Find x .

8. First secant: external 5, whole 20. Second: external 8, whole x . Find x .

9. First secant: external 3, internal 7. Second: external 5, whole x . Find x .

10. A tangent and secant share an external point. Tangent 10; secant external 4, whole 25. Verify the relationship.

11. A tangent has length 15. A secant has external part 9 and whole length x . Find x .

12. A tangent has length 12. A secant has external part 6 and internal part x . Find x .

Day 18 Problem Set

Continued | Mixed Algebra and Error Analysis

13.	Two secants share an external point. First has external 2 and internal 10. Second has external 3 and internal x . Find x .
-----	--

14.	A tangent has length 18. A secant has external part 9 and whole length x . Find x .
-----	---

15.	Intersecting chord parts are $x+1$ and 8 on one chord and 6 and 12 on the other. Find x .
-----	---

16.	Two tangents from the same point measure $7x-4$ and $5x+12$. Find x and the tangent length.
-----	--

17.	Error analysis: A student uses external times internal for a secant. Explain why the rule requires external times whole.
-----	--

18.	Error analysis: A student writes tangent = external times whole. Correct the formula.
-----	---

19.	Spiral review: A central angle and an inscribed angle intercept the same 140-degree arc. Find both angles.
-----	--

20.	Challenge: A secant has external part 4 and internal part 12. A tangent shares its external point. Find the tangent length.
-----	---